

Extended essay: commentary for example essay			
DP subject: (Subject-focused essay)	Biology		
DP subjects: (Interdisciplinary essay)		Interdisciplinary framework:	
Research question:	An investigation into the effect of alcohol production In different yeast strains during the fermentation of wine.		
Assessment details			
Criterion:	Mark awarded:	Commentary: (Incorporating wording from the assessment criteria)	
A: Framework for the essay [Maximum possible mark: 6]	4	The research question is relevant to the topic investigated but it is not clear how the alcohol percentage is measured. The reseach methods are partially suitable but does not justify the choice of measurement of the specific gravity as the dependent variable. Structural conventions used in the essay support communication although raw data should have been presented in an annex instead of the body of the essay.	
B: Knowledge and understanding [Maximum possible mark: 6]	4	Relevant sources have been used to establish knowledge of fermentation but in a rather superficial manner. Relevant terminology is used accurately to demonstrate understanding but there is some terminology that has not been defined. Mentions the use of different "genus" of yeast when they are referring to "species". At times they call them "variety" of yeast. Some relevant concepts are explained and used with partial effectiveness to demonstrate understanding. The relationship between specific gravity and amount of fermentable sugars is not that clear.	
C: Analysis and line of argument [Maximum possible mark: 6]	4	The analysis is partially effective and produces some relevant findings. It is not clear how the specific gravity is turned into percentage alcohol. A t-test was done to find the difference between specific gravity after 7 days fermentation for statistical analysis. A partially consistent line of argument is followed, linking the research question with the findings and conclusion.	
D: Discussion and evaluation [Maximum possible mark: 8]	6	The discussion is balanced, including references to other experiments, but at times in a rather superficial manner. Evaluates different circumstances that might have affected the results, with relevant strengths and limitations described.	
E: Reflection	3	Some evaluation of learning experiences, especially in the finding of a suitable yeast species.	

[Maximum possible mark: 4]		Reflections sometimes shows evidence of growth, specifically in the use of statistical tests.
Total marks awarded	21/30	
Holistic comment on the essay:	The student demonstrates appropriate research skills, resulting in a research question that can be explored within the scope of the effect of different yeast species on the production of alcohol in wine fermentation. There is reasonable engagement with relevant research areas, methods and sources. The relationship between the dependent variable (specific gravity) and the percentage alcohol present is not clear, affecting the outcome of the essay. A more detailed description of how this conversion is obtained could have given a better framework for the discussion of the results. There is good knowledge and understanding of yeast fermentation in the wider context of the subject and there is a reasonably effective application of source material and correct use of subject-specific terminology and concepts. Conclusions are consistent, relevant but analysed in a rather superficial manner. Arguments show reason and are supported by evidence from appropriate scientific papers, but there is very little on the effect of different species of yeast. Research is critically evaluated, with strengths and limitations being presented. Coherent and consistent structure support readability of the essay, with some flaws in presentation. Reflection on the extended essay learning experience is partially evaluative and includes a few specific examples of challenges presented. The reflection shows some evidence of the student's growth and transfer of learning.	